AN INTEGRATED SMART SYSTEM FOR CASHEW CULTIVATION

Project ID - 25-26J-175

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

DEVELOPMENT OF CASHEW LEAF DESEASE DETECTION AI MODEL

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

27/8/25 ..... 27/08/2025

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ABSTRACT

Cashew nut cultivation is a significant agricultural sector in Sri Lanka and many tropical countries, contributing substantially to rural economies. However, its yield is severely threatened by various foliar diseases, which, if not identified and treated early, can lead to catastrophic crop losses. Current disease identification methods are slow, relying on the manual scouting of experts, which is not scalable, often inaccurate, and inaccessible to smallholder farmers. This delay between infection and treatment is a primary cause of reduced productivity.

This research addresses this critical gap by proposing the development of an AI-driven mobile application for the real-time detection and classification of cashew leaf diseases. The core of the system is a two-stage Convolutional Neural Network (CNN) model. The first stage acts as a binary classifier to distinguish between healthy and diseased leaves. If a disease is detected, the second stage, a multi-class classifier, identifies the specific disease from a predefined set (e.g., Anthracnose, Leaf Miner, Red Rust, Leaf Spot). The model will be specifically designed and optimized for deployment on mobile devices, allowing farmers to simply capture an image of a leaf using their smartphone camera and receive an immediate diagnosis.

The project will involve the creation of a curated dataset of cashew leaf images, the design and training of the lightweight CNN models using transfer learning techniques, and the development of a user-friendly Android application. The research will rigorously evaluate the model's performance based on accuracy, precision, recall, and its operational efficiency on mobile hardware. This solution promises a scalable, cost-effective, and accessible tool for farmers, enabling timely intervention, reducing unnecessary pesticide use, and ultimately securing cashew yields and farmer livelihoods.

Keywords

1. Anthracnose, Leaf Miner, Red Rust, Leaf Spot
2. Foliar disease detection
3. Convolutional Neural Network (CNN)

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
AI	Artificial Intelligence
UI	User Interface
CNN	Convolutional Neural Network

Table 1: List of Abbrevia on

1. INTRODUCTION

Cashew nut cultivation is a vital agricultural industry, serving as a primary source of income and livelihood for millions of farmers in tropical regions across the globe, including Sri Lanka. However, the productivity and economic viability of this sector are consistently undermined by a multitude of foliar diseases. Pathogens and pests, manifesting as spots, lesions, rust, and miner damage on leaves, can drastically reduce photosynthetic capacity, weaken trees, and lead to catastrophic yield losses if not identified and managed promptly [1], [2].

The current paradigm for diagnosing these threats remains rooted in traditional methods: manual visual inspection by agricultural experts or farmers. This approach is fundamentally constrained by its reliance on human expertise, which is often scarce, expensive, and inaccessible to smallholder farmers in remote areas. The process is slow, subjective, and not scalable, leading to critical delays between the onset of an infection and the implementation of control measures [3]. These delays allow diseases to spread unchecked, resulting in the unnecessary overuse of broad-spectrum pesticides, increased production costs, and significant financial hardship for farming communities [4], [6].

In this context, technological intervention is not just beneficial but essential. The convergence of advanced artificial intelligence (AI), particularly in the field of computer vision, with the ubiquitous penetration of smartphone technology, presents an unprecedented opportunity to revolutionize agricultural practices [5], [7], [8]. Deep learning–based models, including CNNs, R-CNNs, and Swin Transformers, have already shown remarkable success in plant disease detection across different crops [4], [8], [9]. These advancements demonstrate that AI-powered approaches can outperform traditional methods in terms of speed, scalability, and diagnostic accuracy.

This research proposes to harness this potential by developing an intelligent, accessible, and real-time solution to the persistent challenge of cashew leaf disease identification. Specifically, the project aims to design, develop, and validate a mobile application powered by a Convolutional Neural Network (CNN) that can provide instant and accurate diagnosis of cashew leaf diseases directly from a smartphone camera [1], [2], [6]. By moving diagnostic capabilities from the laboratory to the field and from the expert to the farmer, this system seeks to bridge a critical gap in agricultural technology, enabling timely interventions, promoting sustainable farming practices, and securing the economic stability of cashew-dependent communities.

The following sections will provide a comprehensive background on the existing research in this field, clearly identify the gaps this project intends to fill, and detail the methodology for creating a practical and impactful solution.

1.1 Background

Agriculture is a cornerstone of many national economies, particularly in developing countries such as Sri Lanka, where it not only contributes significantly to GDP but also sustains the livelihoods of millions of smallholder farmers. Among various agricultural commodities, cash crops like cashew (*Anacardium occidentale*) play a crucial role due to their high market value and potential for both local consumption and export. Cashew is widely cultivated in Sri Lanka's dry and intermediate zones, where it supports rural employment and income generation. However, despite its economic significance, cashew production consistently underperforms compared to its potential, with pests and diseases identified as major limiting factors [1], [2], [6].

Cashew trees are vulnerable to a variety of foliar diseases and pest infestations that affect photosynthesis, growth, and nut yield. Common examples include Anthracnose (*Colletotrichum gloeosporioides*), Red Rust (*Cephaleuros virescens*), Leaf Miner infestations, and Leaf Spot diseases, all of which cause lesions, rust-like growths, discoloration, or premature leaf drop, reducing overall yield quality [1], [2]. Studies highlight that such diseases, if not detected and controlled early, can reduce cashew yields by up to 30–50% [2].

Traditionally, disease identification in cashew cultivation relies on manual field inspections. While useful on a small scale, this method presents major drawbacks. It requires expert knowledge in plant pathology, which is often limited in rural farming communities, and is both labor-intensive and time-consuming [2], [6]. Moreover, human observation is prone to error, especially when symptoms overlap across diseases or appear subtly in early stages [3]. In regions like Sri Lanka, where the ratio of agricultural extension officers to farmers is low, many farmers are left without timely guidance, leading to disease spread and economic losses.

The advent of Artificial Intelligence (AI) and deep learning has transformed plant disease detection, particularly through Convolutional Neural Networks (CNNs). CNNs have proven highly effective for image-based classification tasks, automatically learning hierarchical features without requiring handcrafted descriptors [3], [4]. Their success has been

demonstrated across a wide range of crops, including tomato [4], brinjal [10], coffee [7], and even cashew [1], [2], [6]. Recent research has also explored advanced architectures, such as EfficientNet [1], transformers [8], and hybrid approaches combining attention and multi-scale fusion [8], showing performance levels surpassing 95% accuracy in certain cases. These findings highlight CNNs and related models as promising tools for reliable cashew disease detection.

In addition, there has been increasing interest in applying AI-driven plant disease detection through drones [5] and image segmentation models such as SAM [7], further expanding possibilities for precision agriculture. However, while such methods are powerful, they often rely on cloud-based infrastructures or high-performance computing environments, making them less accessible to smallholder farmers in rural areas.

With the rapid penetration of smartphones into rural communities, mobile-based AI solutions are emerging as a practical alternative. Running models directly on devices ensures fast, offline disease detection without the need for internet connectivity, while also protecting farmers' data privacy [2], [6]. Lightweight CNN architectures such as MobileNet, ShuffleNet, and optimized versions of EfficientNet [1] can be deployed on resource-constrained devices, providing accurate results in real time.

Despite significant progress in plant disease detection research for crops like rice, tomato, and maize [3], studies specifically focusing on cashew remain limited. Cashew leaf diseases present unique challenges, as symptoms vary with environmental conditions and multiple infections often co-occur [2]. Therefore, a specialized AI-based mobile solution designed for cashew is essential. Such a system would empower farmers with immediate diagnostic capabilities, enabling timely interventions, reducing unnecessary pesticide use, and safeguarding both crop yield and rural livelihoods.

1.2 Literature Review

The detection and classification of plant diseases using AI and deep learning has seen significant advancements in recent years, with various studies exploring CNN-based architectures and hybrid models. Liu et al. [1] focused specifically on cashew nut diseases using the EfficientNetV2-S architecture, demonstrating high accuracy in classifying multiple foliar diseases, including Anthracnose and Red Rust. Their study utilized a carefully curated cashew leaf dataset and implemented transfer learning to improve feature extraction while minimizing training time. Despite achieving promising results, the study highlighted limitations related to real-time deployment, as the model was tested primarily in controlled environments rather than in-field scenarios, raising concerns about performance under varying lighting and background conditions. Similarly, Amirthavinayagam and Meenakshi [2] applied deep learning techniques to cashew disease management, emphasizing a multi-class classification approach. They reported improved detection rates using CNN-based architectures but noted challenges with class imbalance and misclassification of visually similar diseases, indicating the need for more diverse and representative datasets.

Broader systematic reviews on plant disease recognition, such as the work by Abade et al. [3], have shown that while CNNs outperform traditional machine learning methods in terms of accuracy and scalability, most studies are limited by small datasets, lack of field variability, and overfitting to laboratory conditions. In crop-specific studies, Priyadharshini and Judie Dolly [4] compared CNN, R-CNN, Fast R-CNN, and Faster R-CNN for tomato leaf disease detection, finding that while Faster R-CNN provided superior detection accuracy and localization of disease spots, it was computationally intensive, limiting real-time applicability on mobile devices. Yehia et al. [5] introduced drone-based disease detection combined with extensive data augmentation techniques, including brightness adjustment, rotation, and Gaussian noise, which improved model robustness under real-world conditions. However, their approach relied heavily on cloud-based processing, creating latency and requiring constant internet access, which is often impractical for rural farmers.

Specific to cashew, research by M. P. R. R. et al. [6] employed machine learning for precise disease classification, achieving reasonable accuracy but still encountering issues with distinguishing early-stage infections and co-occurring diseases. Emerging approaches using advanced image segmentation models, such as the Segment Anything Model (SAM), as demonstrated by Vides-Flores et al. [7], show potential for isolating relevant leaf regions from complex backgrounds, enhancing model focus on disease features. Additionally, Karthik et al. [8] explored the combination of Swin Transformers and dual-attention multi-scale fusion networks for crop disease classification, reporting high performance due to the model's ability to capture both local and global features. While highly accurate, such

models are computationally demanding, which poses challenges for mobile deployment. Studies by Sathya et al. [9], [10] on tomato and brinjal leaf diseases further reinforce that while traditional CNNs can achieve satisfactory accuracy, performance drops when tested in diverse field conditions or with unseen image variations, emphasizing the need for domain-specific fine-tuning and augmentation strategies.

Overall, these studies collectively illustrate the potential of deep learning in plant disease detection but reveal common limitations: many models are either trained on controlled datasets or rely on cloud-based computation, which impedes real-time, offline usability in rural areas. Moreover, the lack of specialized cashew datasets, two-stage classification pipelines, and mobile-friendly implementations highlights a critical research gap. These insights justify the methodological focus of this study on developing a two-stage CNN system, optimized for cashew leaf diseases, capable of offline, real-time detection on mobile devices, and robust under real-world environmental variations.

1.3 Research Gap

Despite the notable progress in applying Artificial Intelligence (AI) to agriculture, several critical gaps remain in developing a practical solution tailored for cashew farmers. One of the most pressing issues is the **lack of specialized cashew models**. Existing research often focuses on generic plant disease detection models, which do not adequately address the unique characteristics of cashew leaves. Variations in lighting conditions, leaf maturity, and regional strains of diseases are rarely considered, leading to lower accuracy when applied in real-world scenarios. A fine-tuned model trained on a comprehensive cashew-specific dataset is essential for achieving reliable detection results.

Another limitation observed in prior studies is the dependence on **cloud-centric solutions**. Many proposed systems require captured images to be uploaded to remote servers for processing. While this approach benefits from high computational resources, it introduces a dependency on internet connectivity, which is often unreliable in rural farming areas where cashew plantations are located. Moreover, this method creates latency in processing, which can hinder real-time disease diagnosis—a critical factor for timely agricultural interventions.

A further gap is related to the **classification methodology** employed in disease detection systems. Several existing solutions rely on single-stage classification, where images are directly categorized into multiple disease classes. This approach is not only computationally intensive but also prone to misclassifying healthy leaves as diseased ones. A two-stage classification pipeline, where the system first distinguishes between healthy and diseased leaves and subsequently identifies the

specific disease, offers higher efficiency and improved accuracy. This approach is particularly suitable for mobile or edge-based implementations.

Lastly, there is a significant challenge in terms of **integration and usability**. Much of the current research primarily focuses on developing accurate models without bridging the gap between the trained models and their deployment in farmer-centric applications. The absence of intuitive and accessible mobile applications prevents these technologies from being effectively adopted by cashew farmers, many of whom may have limited digital literacy. A complete solution must, therefore, go beyond accuracy metrics and incorporate a user-friendly, mobile-first design to ensure real-world usability.

This research aims to address these gaps by developing an integrated, mobile-first, two-stage CNN-based solution specifically designed for cashew leaf diseases. By emphasizing specialized datasets, offline processing capabilities, efficient classification, and farmer-friendly interfaces, the study seeks to deliver a practical and accessible tool for improving disease management in cashew cultivation.

Gap	**Current Limitation in Research**	**Proposed Solution in This Study**
Lack of Specialized Cashew Models	Existing models are generic and fail to account for cashew-specific variations (regional disease strains).	Develop a CNN model fine-tuned on a comprehensive, diverse cashew leaf dataset.
Cloud-Centric Solutions	Reliance on internet connectivity introduces latency and limits usability in rural areas.	Implement mobile-first, offline-capable disease detection to reduce dependency on cloud processing.
Single-Stage Classification	Direct classification into multiple disease categories is computationally heavy and causes misclassification.	Use a two-stage pipeline: (1) Healthy vs. Diseased, (2) Specific Disease classification for efficiency and accuracy.
Integration and Usability	Research stops at model accuracy; lacks farmer-friendly applications.	Build an integrated, intuitive mobile application designed for farmers with low digital literacy.

1.4 Research Problem

Cashew cultivation plays a vital role in the agricultural economy of many regions, including Sri Lanka, but its productivity is often threatened by leaf diseases such as Red Rust and Anthracnose. These diseases significantly reduce both the yield and quality of cashew production if not detected and controlled at an early stage. Traditionally, disease identification relies on manual inspection by farmers or agricultural officers, which is highly subjective, time-consuming, and prone to human error, especially when symptoms appear in subtle or overlapping forms.

Although modern image-processing and artificial intelligence techniques have been applied to plant disease detection in general, there is still a lack of robust, crop-specific solutions designed for cashew leaves. Many existing models are trained on generic plant datasets, leading to reduced accuracy and poor adaptability when applied to cashew-specific diseases under natural field conditions, such as varying lighting, leaf orientation, or background noise. This gap limits farmers' ability to use existing systems effectively.

Therefore, the research problem addressed in this study is the absence of an accurate, efficient, and cashew-specific automated disease detection system that can reliably identify and classify Red Rust and Anthracnose using deep learning methods like Convolutional Neural Networks (CNNs).

1. How can a CNN-based model be developed and trained to accurately detect and classify Red Rust and Anthracnose in cashew leaves?
2. What preprocessing and data augmentation techniques are most effective for improving model performance on cashew leaf images under real-world field conditions?
3. How can the system be deployed in a mobile-friendly, user-centric application that is accessible to farmers with minimal digital literacy?
4. How accurate and reliable is the proposed system when tested with real-world images collected from different cashew plantations?

2. OBJECTIVES

2.1 Main Objectives

To design, develop, and evaluate a mobile application for Android that utilizes an integrated two-stage CNN model to provide real-time, accurate, and offline detection and classification of common cashew leaf diseases from images captured by a smartphone camera.

2.2 Sub Objectives

The sub-objectives of this study are designed to address the research questions and guide the development of a robust cashew leaf disease detection system.

To develop a CNN-based model capable of accurately detecting and classifying Red Rust and Anthracnose in cashew leaves using a comprehensive dataset of healthy and diseased leaf images.

To develop effective preprocessing and data augmentation techniques, including resizing, normalization, rotation, flipping, and scaling, to improve model performance under real-world field conditions such as varying lighting, leaf orientation, and background noise.

To develop a mobile-friendly, user-centric application that allows farmers with minimal digital literacy to easily upload leaf images and receive real-time disease diagnosis.

To develop a validation framework to test the system's accuracy and reliability using real-world images from multiple cashew plantations and optimize the model to ensure consistent performance across diverse environmental conditions.

3. METHODOLOGY

The proposed methodology focuses on developing a computer vision–based system to detect Red Rust and Anthracnose in cashew leaves, two of the most prevalent diseases affecting cashew plantations in Sri Lanka. The methodology is designed to ensure high accuracy, real-world applicability, and farmer-friendly usability. The overall workflow consists of five main stages: data collection, preprocessing, model development, evaluation, and deployment.

3.1 System Architecture

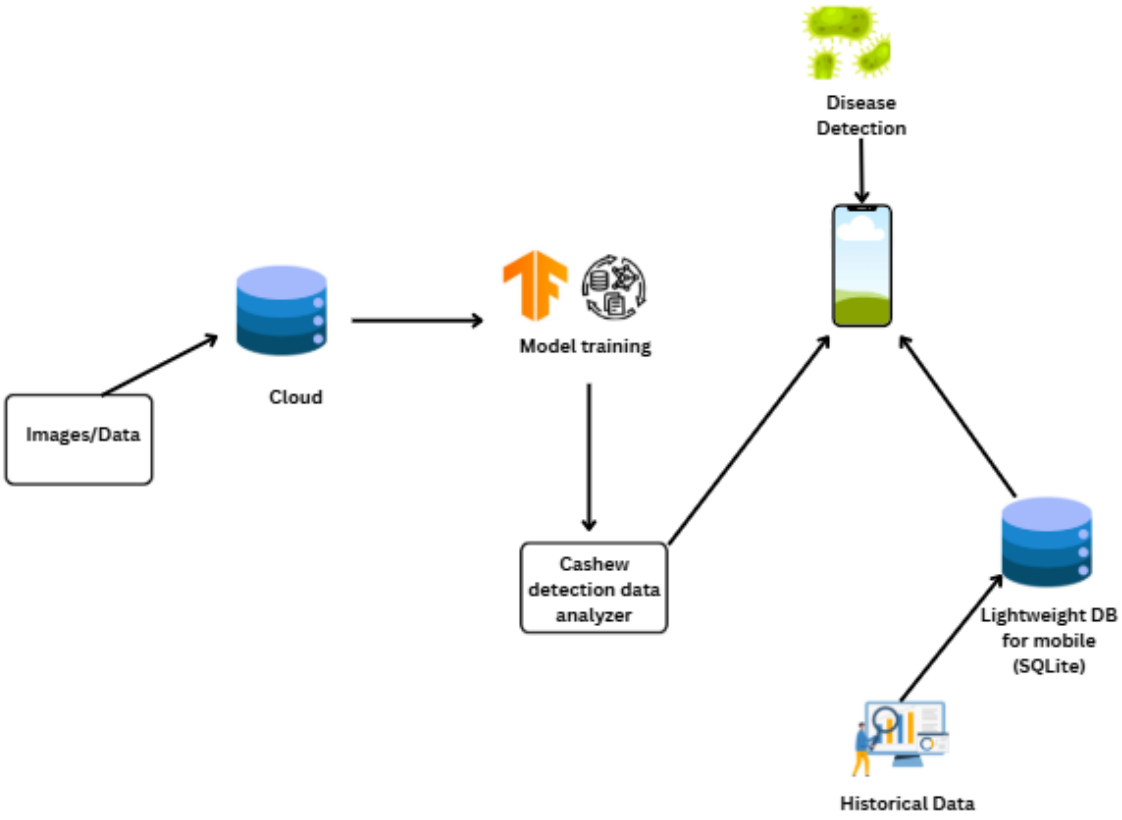

Figure 3.2 Component Architecture Diagram

In this system, images of cashew leaves are first collected and stored in a cloud-based database, which provides scalable storage and accessibility. These stored images are then used for training a machine learning model capable of identifying different diseases affecting cashew leaves. The trained model is integrated into a cashew detection data analyzer and deployed on a mobile device, enabling farmers to capture leaf images and receive real-time disease detection results directly in the field. To ensure smooth operation and offline usability, the mobile application incorporates a lightweight local database (SQLite) for storing user-captured images along with the corresponding detection results. Furthermore, this data is preserved as historical records, allowing farmers to track disease patterns over time and providing valuable feedback for continuous model improvement through retraining in the cloud.

The proposed two-stage CNN-based system for detecting Red Rust and Anthracnose in cashew leaves is justified by its ability to accurately and efficiently address the challenges of real-world disease detection. CNNs are well-suited for capturing the complex visual patterns of leaf diseases, while transfer learning with pretrained models such as ResNet, VGG16, and EfficientNet reduces training time and improves performance on limited datasets. The two-stage classification approach—first distinguishing healthy from diseased leaves, then differentiating between Red Rust and Anthracnose—enhances both accuracy and computational efficiency. Comprehensive preprocessing and data augmentation techniques ensure robustness against variations in lighting, leaf orientation, occlusion, and background clutter. Evaluation using multiple metrics, cross-validation, and real-field testing ensures generalizability and reliability. Furthermore, the model is optimized for mobile deployment with offline capabilities, enabling farmers to capture images and receive real-time disease detection results, track historical patterns, and take timely action. This methodology addresses gaps in previous studies by combining high-performance classification with practical, farmer-friendly usability, making it suitable for improving productivity in Sri Lankan cashew plantations.

Data Collection

High-quality images of cashew leaves will be collected from multiple local plantations in Sri Lanka, covering both healthy leaves and leaves infected with Red Rust and Anthracnose. To ensure the dataset reflects real-world conditions, images will be captured using smartphones and digital cameras under natural lighting. Different leaf orientations, angles, and distances will be recorded to introduce variability that the model will need to generalize across. Environmental factors such as shadows, background clutter, and partial occlusions will also be included to simulate practical field conditions.

To further enhance dataset diversity and mitigate class imbalance, **data augmentation techniques** will be applied. These include:

- **Rotation:** To account for different leaf orientations.
- **Scaling and Cropping:** To ensure the model can detect leaves of varying sizes.
- **Flipping (Horizontal/Vertical):** To introduce symmetry and variability.
- **Brightness and Contrast Adjustment:** To simulate varying lighting conditions throughout the day.
- **Gaussian Noise Addition:** To make the model more robust against background clutter or sensor noise.

Images will be labeled into three categories: Healthy, Red Rust, and Anthracnose, with verification by agricultural experts to ensure accuracy and reduce mislabeling.

Preprocessing

Collected images will undergo several preprocessing steps to prepare them for model training:

- **Resizing:** All images will be standardized to a fixed resolution (e.g., 224×224 pixels) to maintain consistent input dimensions for the CNN.
- **Normalization:** Pixel values will be normalized to a range of 0–1 or standardized with mean and standard deviation to accelerate convergence during training.
- **Noise Reduction:** Background clutter and irrelevant objects will be minimized using filtering techniques such as Gaussian blur or median filtering.
- **Segmentation (Optional)** Techniques like color thresholding or edge detection may be employed to isolate the leaf from its background, enhancing model focus on relevant features.
- **Label Encoding** Labels will be converted into numerical form for supervised learning tasks.

These preprocessing steps ensure that the input data is consistent, clean, and suitable for CNN training, while also addressing the variations found in real-field images.

Model Development:

A **Convolutional Neural Network (CNN)** will serve as the backbone of the disease

detection system, leveraging its proven strength in image classification and pattern recognition.

- **Pretrained Models & Transfer Learning** Architectures such as **ResNet, VGG16, or EfficientNet** will be fine-tuned on the cashew leaf dataset. Transfer learning accelerates training, reduces computational requirements, and improves performance on smaller datasets by leveraging pre-learned features.
- **Hyperparameter Optimization** Key parameters such as learning rate, batch size, number of epochs, optimizer type (e.g., Adam, SGD), and dropout rates will be systematically tuned to maximize performance while preventing overfitting.
- **Two-Stage Classification Approach**
    1. **Stage 1** Detect whether a leaf is healthy or diseased.
    2. **Stage 2** If diseased, classify the leaf into Red Rust or Anthracnose.
       This approach improves computational efficiency and reduces misclassification of healthy leaves.

Evaluation

The trained CNN will be evaluated using multiple metrics to assess performance comprehensively:

- **Accuracy:** Percentage of correctly classified leaf images.
- **Precision, Recall, and F1-Score** To evaluate the model's ability to correctly identify disease classes and handle class imbalance.
- **Confusion Matrix** Visual representation to analyze where the model misclassifies between disease types.
- **Cross-Validation** K-fold cross-validation will be applied to ensure robustness and minimize overfitting, confirming the model's generalizability across unseen data.
- **Real-World Testing** Additional testing will be performed using new images captured under diverse field conditions to validate practical applicability.

Deployment

Once validated, the trained CNN model will be integrated into a **mobile application or portable device** for farmer use:

- **Mobile App Interface:** Farmers can capture or upload leaf images directly from their smartphones.
- **Real-Time Inference:** The CNN processes images on-device or via a lightweight edge computing solution, providing instant disease detection results.
- **User-Friendly Display:** Results will show leaf health status and specific disease name, optionally including recommended action steps.
- **Offline Capability:** The application will be designed to function even in low-connectivity areas, critical for rural plantations.

This methodology ensures that the system is **accurate, practical, and user-friendly**, enabling early disease detection, reducing crop loss, and supporting improved productivity in Sri Lanka's cashew plantations.

3.2 Complete System Architecture Diagram

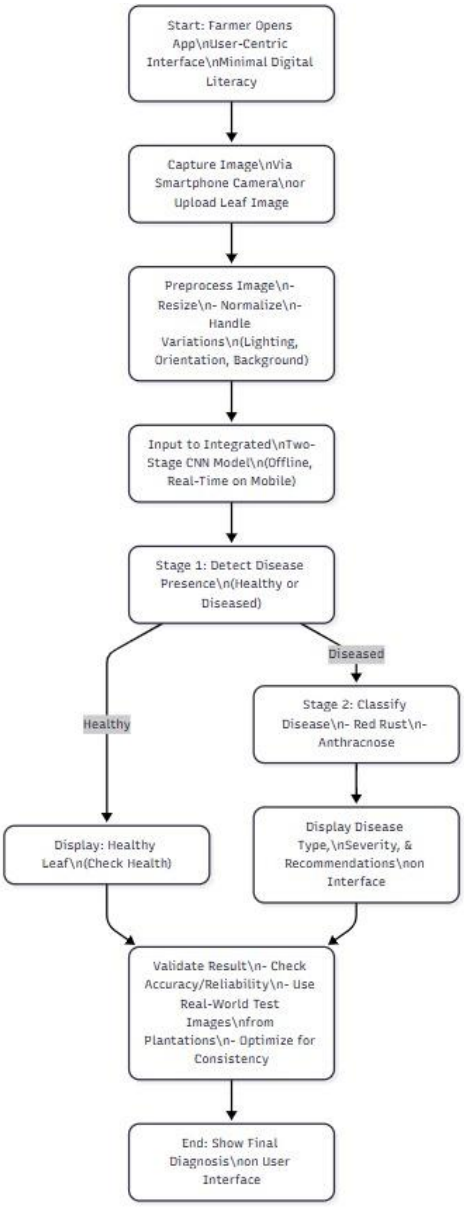

Figure 3.2 Complete System Architecture Diagram

3.3 System Architecture

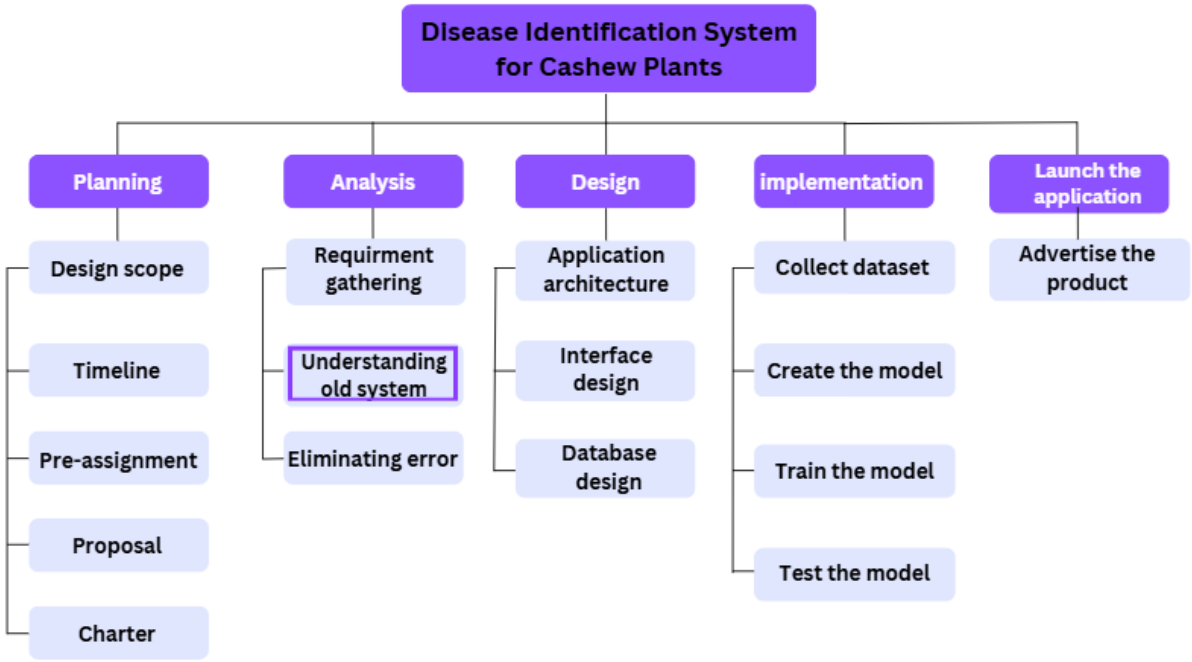

Figure 3.3 Work breakdown structure

3.4 Gnatt Chart

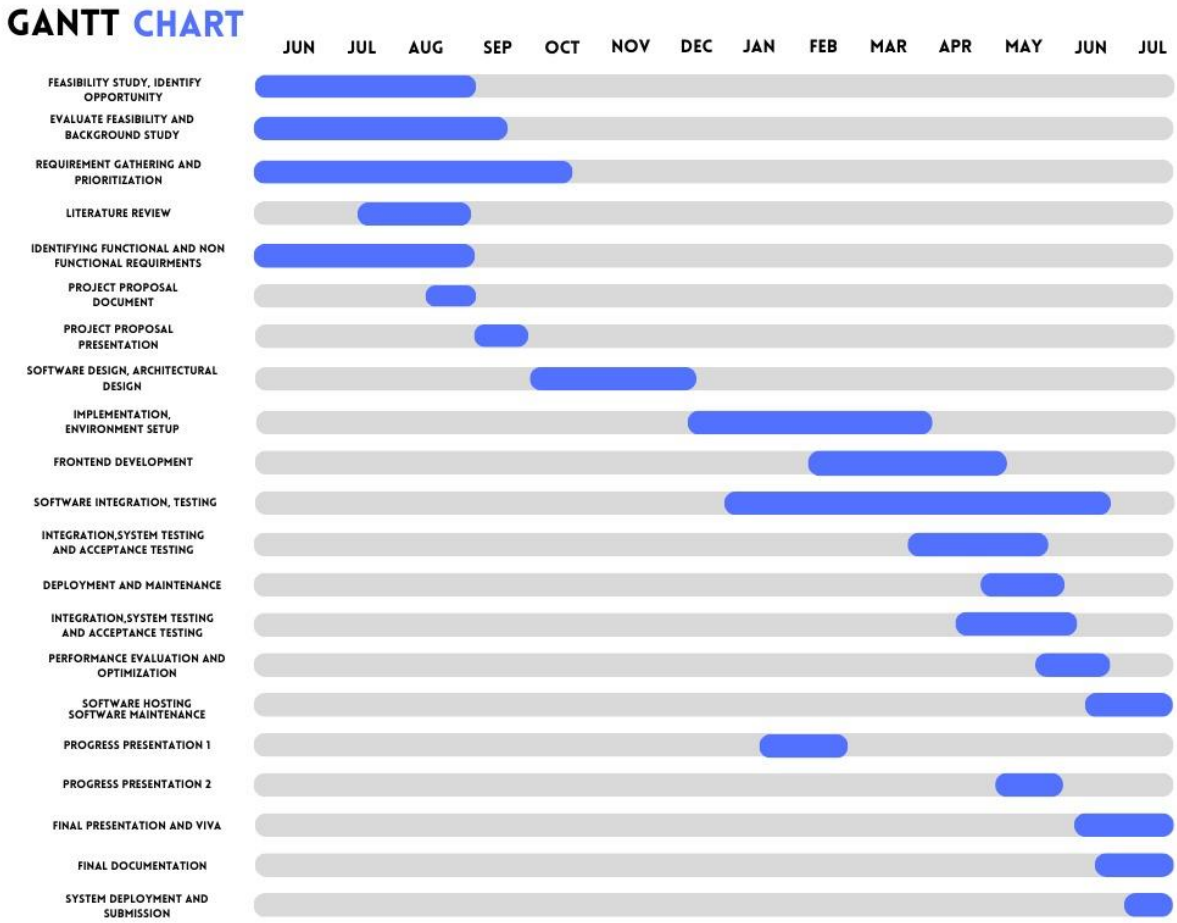

Figure 3.4 Gnatt Chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. The system must allow users to capture an image of a cashew() leaf using the device's camera.
2. The system must provide a diagnosis (Healthy or specific Disease name) within a few seconds.
3. The system must function without an active internet connection (offline capability).
4. The system must display information about the identified disease.

4.2 Non-Functional Requirements

1. Accuracy: The model should achieve >90% accuracy on the test set.
2. Performance: The inference time (image capture to result) should be under 5 seconds on a mid-range smartphone.
3. Usability: The user interface must be simple, intuitive, and require minimal training to use.
4. Size: The final application size (including models) should be optimized to be under 50MB.

4.3 Expected Test Cases

1. Test Case: Healthy Leaf Identification

- Description**:** An image of a healthy, disease-free cashew leaf is captured and processed.
- Expected Result**:** The system correctly identifies the leaf as "Healthy". The second-stage disease classification model is not triggered.

2. Test Case: Diseased Leaf Detection and Classification

- Description**:** An image of a cashew leaf with clear visual symptoms of a known disease (e.g., Anthracnose) is captured and processed.
- Expected Result**:** The system correctly identifies the leaf as "Diseased" and successfully triggers the second-stage model to provide the specific classification (e.g., "Anthracnose").

3. Test Case: Unidentified or Complex Disease Symptom

- Description**:** An image of a cashew leaf with severe, mixed, or unknown disease symptoms is captured. This tests the model's robustness.
- Expected Result**:** The system correctly identifies the leaf as "Diseased". The second-stage model either provides its best prediction with a lower confidence score or correctly indicates it cannot classify the specific disease with high certainty (e.g., "Diseased - Unknown Type").

4. Test Case: Non-Leaf Image Rejection

- Description**:** An image not containing a cashew leaf (e.g., a hand, soil, the sky, or a different plant) is uploaded.
- Expected Result**:** The system either fails to find a leaf and returns an error message (e.g., "Please capture a cashew leaf") or correctly identifies the image as "Healthy" with very low confidence, indicating the input is invalid. This tests the model's ability to handle "out-of-distribution" data.

5. Test Case: Mobile Performance and Offline Functionality

- Description**:** The mobile application is used without an internet connection to capture and process multiple leaf images in sequence.
- Expected Result**:** The application functions fully offline. The inference time (from capture to result display) for each image is consistently under 3-5 seconds on a mid-

range smartphone, demonstrating efficient on-device processing. The application does not crash during the batch processing.

4.4 Technology Stack

Purpose	**Technology**
Programming Language	Python (Model) , Kotlin/Java (App)
ML Framework	TensorFlow, Keras
Mobile ML Framework	TensorFlow Lite (Key Technology)
Base Model Architecture	MobileNetV2
Image Processing	OpenCV, TensorFlow Augmentation
Mobile App Development	Android Studio/Flutter
Training Hardware	Google Colab (GPU)
Version Control	Git, GitHub

Table 4.1: Technology Stack

5. CONCLUSION

Cashew farming is a crucial source of income for many communities, but it faces a constant threat from leaf diseases. Right now, finding these diseases relies on experts manually checking trees. This method is too slow, doesn't cover enough ground, and is often out of reach for the average farmer. The resulting delays in treatment lead to smaller harvests and real financial strain on farming families.

This project offers a modern solution to this old problem. We plan to develop a mobile app that uses artificial intelligence (AI) to identify cashew leaf diseases in real-time. At its heart is a smart, two-step image recognition system designed to work efficiently on a standard smartphone. This makes the solution not only powerful but also practical for everyday use.

Our approach tackles the core issues head-on. Instead of a general plant disease app, ours is built specifically for cashew, which means better accuracy. It works entirely on a phone without needing the internet, making it perfect for remote farms. The two-step process—first checking if a leaf is healthy, then identifying the specific disease—makes the app faster and clearer to use. Most importantly, we are designing it with farmers in mind, ensuring the technology is easy to use and truly helpful.

We believe the successful development of this tool will empower farmers. By providing an instant diagnosis in the field, it enables faster, more targeted treatment. This can help stop diseases from spreading, reduce the overuse of pesticides, prevent crop loss, and support more productive and sustainable cashew farms.

In short, this project is more than just a piece of technology, it's a practical tool designed to solve a pressing real-world problem. By putting advanced AI directly into the hands of farmers, this research has the potential to make a meaningful difference in the field of sustainable agriculture and support the livelihoods of those who depend on cashew cultivation. Our next steps will be to build, test, and refine this system to bring these benefits to life.

6. BUDGET AND BUDGET JUSTIFICATION

Item	**Description**	**Cost Estimation**
Play Store	Deploy the app in the Play Store	7,550 LKR
Google Cloud	12-months	9100 LKR
Total		8,650 LKR

Table 6: Budget

7. REFERENCES

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